

- ④ In all examples we had polynomials in both the numerator and denominator of the L.T. and the inverse was combinations of exponentials (including sin/cos).
- ② For a qualitative description of the inverse transform one can just look at the **poles** of the Laplace transform.
- ③ For higher order poles we use:

$$\frac{1}{(s-a)^{k+1}} = \frac{(-1)^k}{k!} \frac{d^k}{ds^k} \left(\frac{1}{s-a} \right) = \mathcal{L}\left[\frac{1}{k!} t^k e^{at}\right]$$

Examples: describe the signal based on the poles of its Laplace transforms

①

$$Y(s) = \frac{3s-1}{s^2+4}$$

②

$$Y(s) = \frac{3s-1}{(s+2i)(s-2i)} \rightarrow y(t) = \alpha e^{2it} + \beta e^{-2it} = \alpha' \cos 2t + \beta' \sin 2t$$

③

$$Y(s) = \frac{s-1}{(s+1)(s-2)} \rightarrow y(t) = \alpha e^{-t} + \beta e^{2t}$$

$$H(s) = \frac{a}{s^2(s^2+a^2)}$$

$$H(s) = \frac{a}{s^2(s+ai)(s-ai)} \rightarrow y(t) = \alpha t + \beta \sin(at) + \gamma \cos(at)$$

Remember!

- ① Poles corresponds to different signal components which appear additively in the signal.
- ② Poles of order $k > 1$ correspond to the same type of signal as the simple pole ($k = 1$) multiplied by t^{k-1} .

Laplace transform and differential equations

Let us solve the following linear constant coefficient inhomogeneous differential equation:

$$y'' - 4y = g(t), \quad y(0) = 3, \quad y'(0) = -6$$

Usual way:

Solve the homogeneous equation: $y_h'' - 4y_h = 0$

Look for the solution in the form

$$\rightarrow y_h(t) = e^{rt} \rightarrow r^2 - 4 = 0$$

$$\rightarrow r = \pm 2$$

$$\rightarrow y_{1,2} = e^{\pm 2t}$$

$$\rightarrow y_h = C_1 e^{2t} + C_2 e^{-2t}$$

Particular solution:

Get the particular solution (satisfying the orig. eq.) by variation of constants:

$$y_p = u_1(t)y_1 + u_2(t)y_2 = u_1(t)e^{2t} + u_2(t)e^{-2t}.$$

To avoid second derivatives of u_1, u_2 , impose

$$u_1' e^{2t} + u_2' e^{-2t} = 0.$$

Substituting into $y'' - 4y = g(t)$ gives

$$2u_1' e^{2t} - 2u_2' e^{-2t} = g(t).$$

Thus

$$y(t) = 3e^{-2t} + \frac{1}{4} \int_0^t \left(e^{2(t-\tau)} - e^{-2(t-\tau)} \right) g(\tau) d\tau$$

Thus

$$\begin{cases} u_1' e^{2t} + u_2' e^{-2t} = 0, \\ 2u_1' e^{2t} - 2u_2' e^{-2t} = g(t). \end{cases}$$

$$u_1' = \frac{1}{4} g(t) e^{-2t}, \quad u_2' = -\frac{1}{4} g(t) e^{2t}.$$

$$u_1(t) = \frac{1}{4} \int_0^t g(\tau) e^{-2\tau} d\tau,$$

$$u_2(t) = -\frac{1}{4} \int_0^t g(\tau) e^{2\tau} d\tau.$$

$$y_p(t) = \frac{1}{4} e^{2t} \int_0^t g(\tau) e^{-2\tau} d\tau - \frac{1}{4} e^{-2t} \int_0^t g(\tau) e^{2\tau} d\tau.$$

The full solution is

$$y(t) = C_1 e^{2t} + C_2 e^{-2t} + y_p(t).$$

The initial conditions give

$$C_1 = 0, \quad C_2 = 3.$$

Redo it using the Laplace transform:

$$y'' - 4y = g(t), \quad y(0) = 3, \quad y'(0) = -6 \quad | \mathcal{L}[\dots]$$

$$\mathcal{L}[y'] = sY(s) - y(0), \quad \mathcal{L}[y''] = s^2 Y(s) - sy(0) - y'(0), \quad \mathcal{L}[g] = G(s).$$

Apply the Laplace transform on both sides:

$$s^2 Y(s) - 3s + 6 - 4Y(s) = G(s).$$

Hence

$$(s^2 - 4)Y(s) = G(s) + 3s - 6.$$

Therefore

$$Y(s) = \frac{3s - 6}{s^2 - 4} + \frac{G(s)}{s^2 - 4}.$$

We split it into standard inverse-transform terms:

$$Y(s) = 3 \frac{s}{s^2 - 4} - 3 \frac{2}{s^2 - 4} + \frac{1}{2} \frac{2}{s^2 - 4} G(s).$$

Using

$$\mathcal{L}^{-1} \left[\frac{s}{s^2 - 4} \right] = \cosh 2t, \quad \mathcal{L}^{-1} \left[\frac{2}{s^2 - 4} \right] = \sinh 2t,$$

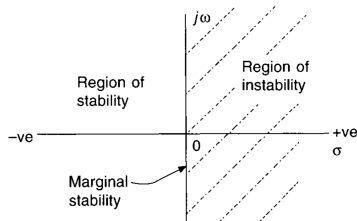
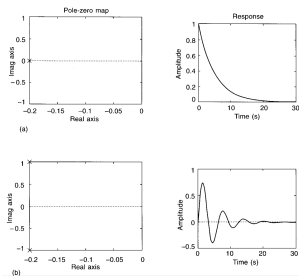
and the convolution theorem, we get

$$y(t) = \underbrace{3e^{-2t}}_{\text{transient}} + \underbrace{\frac{1}{2} \int_0^t \sinh(2(t - \tau)) g(\tau) d\tau}_{\text{steadystate}}$$

If the pole of a signal's Laplace transform...

- 1 has a negative/positive real component, σ , $\rightarrow$ exponential decay/growth with rate σ .
- 2 has imaginary components $\pm i\omega \rightarrow$ oscillations with frequency ω .
- 3 lie on the imaginary axis $\rightarrow$ constant amplitude oscillation
- 4 is in the origin $\rightarrow$ step function, i.e., DC (direct current = non-oscillating) signal

Exponentially increasing signals are produced by unstable systems. These must be redesigned!



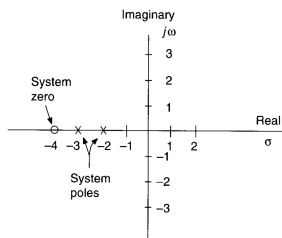
Unlike poles **zeros** do not affect the shape of the signals only their weightings. The nearer a zero is to a pole the less the contribution made by that pole to the signal.

Poles, zeros and $p - z$ diagrams in the s -plane

The shape of the impulse response can be inferred from the properties of the transfer function.

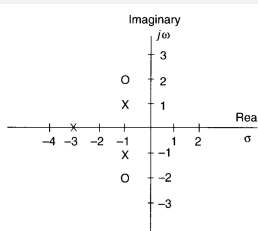
Example

$$H(s) = \frac{s + 4}{(s + 2)(s + 3)}$$



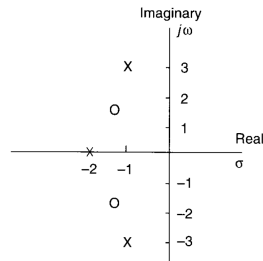
Example

$$H(s) = \frac{s^2 + 2s + 5}{(s + 3)(s^2 + 2s + 2)}$$
$$= \frac{(s + 1 + i2)(s + 1 - i2)}{(s + 3)(s + 1 + i)(s + 1 - i)}$$



Example

$$Y(s) = \frac{3s^2 + 8s + 14}{(s + 2)(s^2 + 2s + 10)}$$
$$= \frac{1}{s + 2} + \frac{2(s + 1)}{s^2 + 2s + 10}$$
$$y(t) = e^{-2t} + 2e^{-t} \cos 3t$$
$$Y(s) = 3 \frac{(s + 1.33 + i1.7)(s + 1.33 - i1.7)}{(s + 2)(s + 1 + i3)(s + 1 - i3)}$$



LTI systems for continuous signals

If we are interested in the **steady state behavior** of the system the most general form of its functioning is:

$$y(t) = \mathcal{H}[x(t)] = \int_0^t x(\tau)h(t - \tau)d\tau = (x * h)(t) .$$

For describing transients one should look into the homogeneous solution of the diff. eq. defining the system.

- ① $h(t)$ is the **impulse response** of the system because for $x(t) = \delta(t)$ (impulse signal) we get $y(t) = h(t)$.
- ② all involved signals are **causal**, i.e. i.e., $x(t) = h(t) = 0$ for $t < 0$,
- ③ steady state $\rightarrow t$ is much larger than the typical timescales of $x(t)$ (e.g. dominant frequency) and $h(t)$ (e.g. resonance freq. of an LC filter).
- ④ time invariance is obvious for positive shifts.
- ⑤ for negative shifts we need to assume that the impulse response has a finite support $\rightarrow$ the output will be shifted as the input apart from an initial transient.

the steady state output of any LTI can be obtained by convolving the input with the impulse response.

Using the Laplace-transform's properties we get: $Y(s) = X(s) \cdot H(s) \rightarrow$

Transfer function

$$H(s) = \mathcal{L}[h(t)] = \frac{Y(s)}{X(s)}$$

The frequency response of continuous systems

We are interested in stable systems and the steady state oscillations they produce when provide oscillatory input, $\rightarrow$ we ignore transients $\rightarrow$ we look at the behavior of the output at times that are much larger than some characteristic time of the system.

Let's see the output $y(t)$ if the the input is a unit amplitude harmonic oscillation: $x(t) = e^{i\omega t}$

$$y(t) = (h * x)(t) = \int_0^t e^{i\omega(t-\tau)} h(\tau) d\tau = e^{i\omega t} \underbrace{\int_0^t e^{-i\omega\tau} h(\tau) d\tau}_{\approx H(i\omega) \text{ for large } t} = e^{i\omega t} H(i\omega)$$

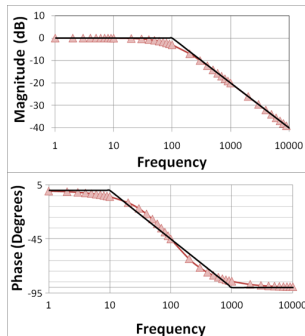
$$H(i\omega) = |H(i\omega)|e^{i\theta(\omega)} \quad \rightarrow \quad y(t) = |H(i\omega)|e^{i(\omega t + \theta(\omega))}$$

Code diagram/plot

For representing the **frequency response of LTI systems**.

- ① **gain** of the system: $|H(i\omega)|$
- ② **phase relationship** output vs input: $\angle H(i\omega) = \theta(\omega)$.

Gain (magnitude) is represented in dB units, frequency in (rad/sec) units (log scale), phase in degrees (linear scale).



Example

Find the gain and phase angle for angular frequencies 0 rad/s (d.c.), 2 rad/s and for very large frequencies if the system is characterised by the transfer function

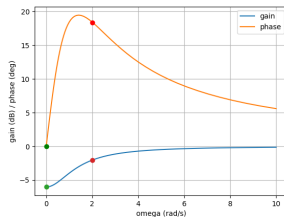
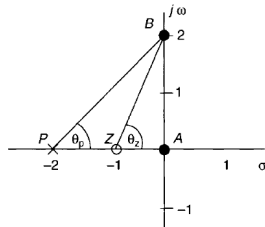
$$H(s) = \frac{s+1}{s+2} \rightarrow |H(i\omega)| = \sqrt{\frac{\omega^2+1}{\omega^2+4}}, \quad \angle H(i\omega) = \arctan \frac{\omega}{1} - \arctan \frac{\omega}{2}$$

$\omega = 0 \text{ rad/s} \rightarrow \text{Gain} = 0.5, \text{ phase angle} = 0^\circ$

$\omega = 2 \text{ rad/s} \rightarrow \text{Gain} = \sqrt{\frac{5}{8}}, \text{ phase angle} = \arctan 2 - \arctan 1 = 18.4^\circ$

$\omega \rightarrow \infty \text{ rad/s} \rightarrow \text{Gain} \rightarrow 1, \text{ phase angle} = \frac{\pi}{2} - \frac{\pi}{2} = 0^\circ$

```
1 from pylab import *
2 from scipy import signal
3 num, den = [1, 1], [1, 2]
4 o, gain, phase = signal.bode((num, den))
5 plot(o, gain) # o is omega
6 plot(o, 20*log10(sqrt(o**2+1)/sqrt(o**2+4)))
7 plot(o, phase)
8 plot(o, (arctan(o) - arctan(o/2))*180/pi)
```



All-pole filters have transfer functions that only have poles and no zeros, i.e. they are of type $H(s) = \frac{1}{P(s)}$ where $P(s)$ is a polynomial.

Exercise

- 1 Explain why all-pole analogue filters can only be low-pass or band-pass.
- 2 What kind of filter does the transfer function $H(s) = \frac{s}{s+1}$ represent? Argue it using words instead of calculations.

Function representation of sequences

- Functions are characterized in many ways (roots, singularities, asymptotic behavior, etc.).
- They can often be uniquely described as linear combinations of linearly independent well-known simple functions. E.g.:
 - Taylor-series of analytic functions using power functions;
 - Fourier series of periodic functions using harmonic functions;
 - Fourier series of square integrable functions using families of orthogonal polynomials such as Hermite or Legendre.

We shall do it the other way around

we shall characterize a set of numbers (our sequence) by the properties of the function that can be constructed via linear combination of a family of simple functions using the sequence as coefficients.

Polynomial representation of sequences

If we look at sequences $a[n]$ and $b[n]$ as coefficients a_n and b_n of corresponding polynomials

$$A_p(x) = \sum_{n=0}^p a_n x^n, \quad B_q(x) = \sum_{m=0}^q b_m x^m$$

then the product of these polynomials

$$C_{p+q}(x) \equiv A_p(x) \cdot B_q(x) = \sum_{n=0}^p \sum_{m=0}^q a_n b_m x^{n+m}$$

$$k \equiv n + m \rightarrow m = k - n \rightarrow C_{p+q}(x) = \sum_{k=0}^{p+q} \sum_{n=0}^p a_n b_{k-n} x^k = \sum_{k=0}^{p+q} c_k x^k$$

corresponds to the convolution sum of the two sequences.

$$c_k = \sum_{n=0}^p a_n b_{k-n} \rightarrow c[k] = a[k] * b[k]$$

Example

Let $a[n] = \{1, 2, 3, 4, 5\}$ and $b[n] = \{-1, 2, 1\}$

$$A(x) = 1 + 2x^2 + 3x^3 + 4x^4 + 5x^5, \quad B(x) = -1 + 2x + 3x^2$$

(1	+	2x	+	3x ²	+	4x ³	+	5x ⁴)	·	(-1	+	2x	+	x ²)
-1	+	-2x	+	-3x ²	+	-4x ³	+	-5x ⁴						
		2x	+	4x ²	+	6x ³	+	8x ⁴	+	10x ⁵				
				x ²	+	2x ³	+	3x ⁴	+	4x ⁵	+	5x ⁶		
=	-	1		+	2x ²	+	4x ³	+	6x ⁴	+	14x ⁵	+	5x ⁶	

$$a[n] * b[n] = \{-1, 0, 2, 4, 6, 14, 5\}$$

The z-transform

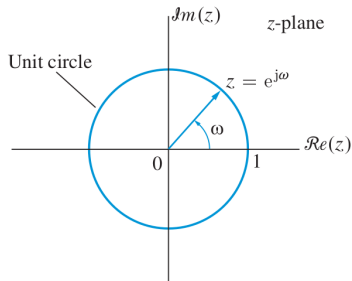
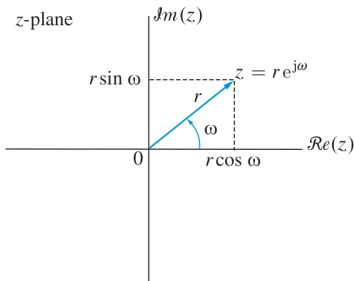
Definition

The z-transform of a sequence $x[n]$ is

$$X(z) \equiv \sum_{n=-\infty}^{\infty} x[n]z^{-n}, \quad z \in \mathbb{C}, \quad X: \mathbb{C} \rightarrow \mathbb{C}.$$

$$\mathcal{O}_x(z^n) \equiv x[n] * z^n = \sum_{k=-\infty}^{\infty} x[k]z^{n-k} = z^n \cdot X(z).$$

→ z^n is the eigenvector of the (linear) convolution operator $\mathcal{O}_x$ with $X(z)$ the corresponding eigenvalue.



- ① The z-transform defines for each sequence a continuous complex-valued surface over the complex plane $\mathbb{C}$.
- ② For finite sequences, its value is always defined across the entire complex plane.
- ③ For infinite sequences, it can be shown that the z-transform converges only for the region

$$\lim_{n \rightarrow \infty} \left| \frac{x[n+1]}{x[n]} \right| < |z| < \lim_{n \rightarrow -\infty} \left| \frac{x[n+1]}{x[n]} \right|$$

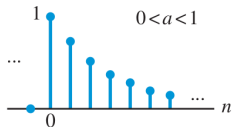
- ④ The z-transform identifies a sequence unambiguously only in conjunction with a given **region of convergence (ROC)**. In other words, there exist different sequences, that have the same expression as their z-transform, but that converge for different amplitudes of z .
- ⑤ The z-transform of a sequence is a generalization of the Fourier transform of the corresponding sampled signal. The latter is restricted to the unit circle ($z = e^{i\omega t} \rightarrow |z| = 1$)

$$\hat{x}(t) = x(t) \cdot \Pi_T(t) \rightarrow \mathcal{F}[\hat{x}](\omega) = T \sum_{n=-\infty}^{\infty} x[n] e^{-i\omega n} \equiv T \cdot X(e^{i\omega}), \quad \omega \equiv 2\pi \frac{f}{f_s}$$

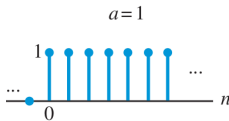
Example: Causal exponential sequence

$$x[n] = a^n u[n] \rightarrow X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a} \rightarrow \text{ROC} : |z| > |a|.$$

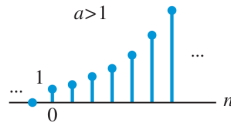
The series converges if $|az^{-1}| < 1 \rightarrow |z| > |a|$. In this case we have a zero at $z = 0$ and a pole at $z = a$.



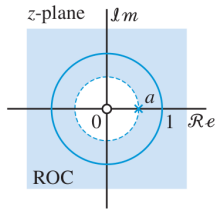
Decaying exponential



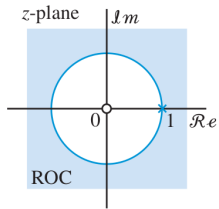
Unit step



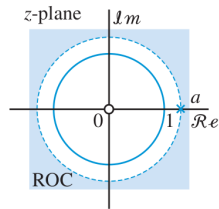
Growing exponential



(a)



(b)

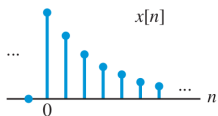


(c)

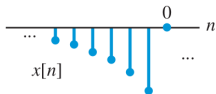
Example: Two sided exponential sequence

$$x[n] = \begin{cases} a^n, & n \geq 0 \\ -b^n, & n < 0 \end{cases} \rightarrow X(z) = - \sum_{n=-\infty}^{-1} (bz^{-1})^n + \sum_{n=0}^{\infty} (az^{-1})^n = \sum_{n=1}^{\infty} \left(\frac{z}{b}\right)^n + \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n.$$

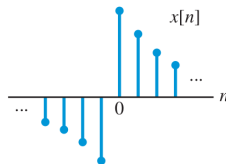
For both terms (anti-causal & causal) to converge: $|a| < |z| < |b|$.



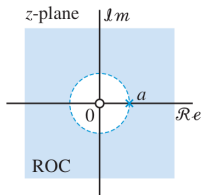
Causal sequence



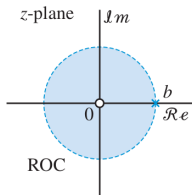
Anticausal sequence



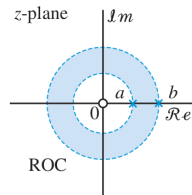
Two-sided sequence



(a)



(b)



(c)

Exercises

Calculate the following z-transforms:

- ① unit step $u[n] = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases} \rightarrow U(z) = \frac{z}{z-1}$
- ② impulse sequence $\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \rightarrow \Delta(z) = 1$

Exercises

Show that if sampled at intervals T the z-transforms of the following

- $f(t)$ functions are ... a.) $t \rightarrow \frac{Tz}{(z-1)^2}$; b.) $t^2 \rightarrow \frac{T^2 z(z+1)}{(z-1)^3}$;
c.) $e^{-at} \rightarrow \frac{z}{z-e^{-aT}}$; d.) $te^{-at} \rightarrow \frac{Tze^{-aT}}{(z-e^{-aT})^2}$;
e.) $\sin \omega t \rightarrow \frac{z \sin \omega T}{z^2 - 2z \cos \omega T + 1}$; e.) $\cos \omega t \rightarrow \frac{z(z - \cos \omega T)}{z^2 - 2z \cos \omega T + 1}$;

Exercise

Calculate the z-transform of the following sequences and determine their region of convergence:

	Sequence $x[n]$	z-Transform $X(z)$	ROC
1.	$\delta[n]$	1	All z
2.	$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
3.	$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
4.	$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
5.	$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
6.	$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
7.	$(\cos \omega_0 n) u[n]$	$\frac{1 - (\cos \omega_0)z^{-1}}{1 - 2(\cos \omega_0)z^{-1} + z^{-2}}$	$ z > 1$
8.	$(\sin \omega_0 n) u[n]$	$\frac{(\sin \omega_0)z^{-1}}{1 - 2(\cos \omega_0)z^{-1} + z^{-2}}$	$ z > 1$
9.	$(r^n \cos \omega_0 n) u[n]$	$\frac{1 - (r \cos \omega_0)z^{-1}}{1 - 2(r \cos \omega_0)z^{-1} + r^2 z^{-2}}$	$ z > r$
10.	$(r^n \sin \omega_0 n) u[n]$	$\frac{(\sin \omega_0)z^{-1}}{1 - 2(r \cos \omega_0)z^{-1} + r^2 z^{-2}}$	$ z > r$

Recap

$$\rightarrow x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k] .$$

Any signal can be decomposed into the superposition of scaled and delayed impulses.

$$u[n] = \delta[n] + \delta[n - 1] + \delta[n - 2] + \dots = \sum_{k=0}^{\infty} \delta[n - k] = \sum_{k=-\infty}^n \delta[k] .$$

We can determine the output of any LTI system if we know its impulse response:

$$\left. \begin{array}{l} \delta[n] \xrightarrow{\mathcal{H}} h[n] \quad \text{impulse response} \\ \mathcal{H} \text{ time-invariant} \rightarrow \delta[n - k] \xrightarrow{\mathcal{H}} h[n - k] \\ x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k] \quad | \mathcal{H}(\dots) \end{array} \right\} \xrightarrow{\mathcal{H} \text{ linear}} x[n] \xrightarrow{\mathcal{H}} y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

Taking the polynomial representations (z-transforms) of the $x[k]$ input and the impulse response sequence $h[k]$ as $X(z)$ and respectively $H(z)$ the response $y[k]$ of the LTI to the former will be:

$$Y(z) = H(z) \cdot X(z) .$$

Transfer function

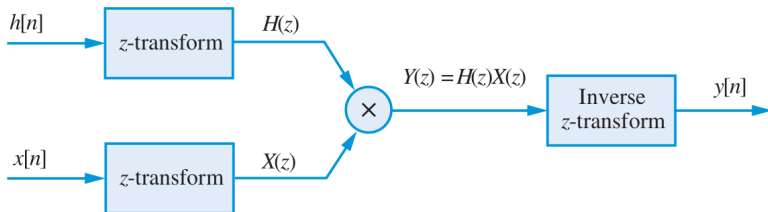
If there is an LTI system transforming the input $x[n]$ with z-transform $X(z)$ into the output $y[n]$ with z-transform $Y(z)$ such that

$$y[n] = x[n] * h[n]$$

the z-transform of the impulse response sequence $h[n]$ is

$$H(z) = \frac{Y(z)}{X(z)}$$

Finding the output of a discrete system using the z-transform:



When applying the inverse z-transform the ROC should be specified along with the expression for the function. Otherwise the underlying sequence is not uniquely defined.

Example

Determine the inverse of

$$H(z) = \frac{z}{z - 0.5} .$$

- For $|z| < 0.5$

$$H(z) = -2z \frac{1}{1 - 2z} = -2z \sum_{n=0}^{\infty} 2^n z^n = - \sum_{n=-\infty}^{-1} 2^{-n} z^{-n} , \quad h[n] = \{ \dots, -8, -4, -2, 0, 0, 0, \dots \} \quad (\text{anti-causal})$$

- For $|z| > 0.5$

$$H(z) = \frac{1}{1 - 0.5z^{-1}} = \sum_{n=0}^{\infty} 0.5^n z^{-n} , \quad h[n] = \{ \dots, 0, 0, 0, 1, 0.5, 0.25, 0.125, \dots \} \quad (\text{causal})$$

Exercise

- 1 If the input to a system, having a transfer function of $1 + 2z^{-1} - z^{-2}$ is a discrete unit step, show that the first four terms of the output sequence is 1, 3, 2, 2.
- 2 When a discrete signal 1, -2 is inputted to a processing system the output sequence is 1, -5, 8, -4.
 - (a) Derive the transfer function for the system.
 - (b) Find the first three terms of the output sequence in response to the finite input sequence of 2, 2, 1.
- 3 A processor has a transfer function $0.4/(z - 0.5)$. Find the response to a discrete unit step (first four terms only).

z-transform of time shift

If a sequence has a z-transform $X(z)$ then the unit shifted sequence's z-transform is $X(z)z^{-1}$.

Exercise

Prove the above.

Exercise

Prove the following properties of the z-transform:

	Property	Sequence	Transform	ROC
		$x[n]$	$X(z)$	R_x
		$x_1[n]$	$X_1(z)$	R_{x_1}
		$x_2[n]$	$X_2(z)$	R_{x_2}
1.	Linearity	$a_1x_1[n] + a_2x_2[n]$	$a_1X_1(z) + a_2X_2(z)$	At least $R_{x_1} \cap R_{x_2}$
2.	Time shifting	$x[n - k]$	$z^{-k}X(z)$	R_x except $z = 0$ or ∞
3.	Scaling	$a^n x[n]$	$X(a^{-1}z)$	$ a R_x$
4.	Differentiation	$nx[n]$	$-z \frac{dX(z)}{dz}$	R_x
5.	Conjugation	$x^*[n]$	$X^*(z^*)$	R_x
6.	Real-part	$\text{Re}\{x[n]\}$	$\frac{1}{2}[X(z) + X^*(z^*)]$	At least R_x
7.	Imaginary part	$\text{Im}\{x[n]\}$	$\frac{1}{2j}[X(z) - X^*(z^*)]$	At least R_x
8.	Folding	$x[-n]$	$X(1/z)$	$1/R_x$
9.	Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	At least $R_{x_1} \cap R_{x_2}$
10.	Initial-value theorem	$x[n] = 0 \text{ for } n < 0$	$x[0] = \lim_{z \rightarrow \infty} X(z)$	